Q $T(2^k) = 3T(2^{k-1}) + 1$

Q $T(0) = 1$

$T(1) = 2$

$T(n) = 5T(n-1) - 6T(n-2)$

$T(2^k) = 3T(2^{k-1}) + 1$

$\frac{2^k}{2}$

$m = 2^k$

$T(m) = 3T(m/2) + 1$

$a = 3$

$b = 2$

$n = 0$

$\log_2^3 > n$

2^{k-1}

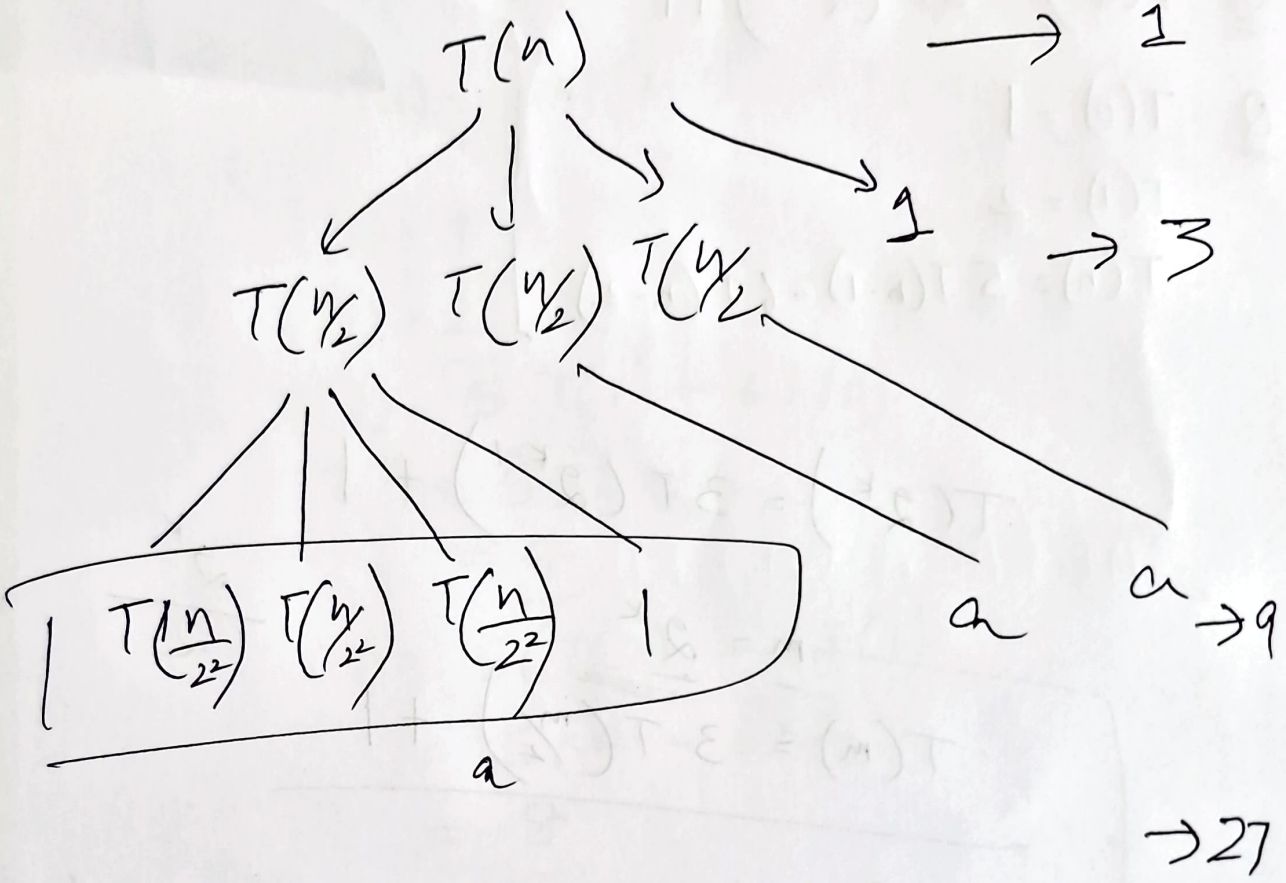
$T\left(\frac{2^k}{2}\right)$

$m^{\log_2^3}$

$2^{k \log_2^3}$

$2^{\log_2^3 k}$

$O(3^k)$


T(n)
1
1
T(n/2) T(n/2) T(n/2)
3
1 T(n/2²) T(n/2²) T(n/2²) 1
a
a
a
9
a
27

Q $T(0) = 1$

$T(1) = 2$

$T(n) = 5\,T(n-1) - 6\,T(n-2) + 0$

$T(0) = 1$

$T(1) = 2$

$T(2) = 5\,T(1) - 6\,T(0)$

$= 5 \times 2 - 6$

$= 4$

$T(3) = 5(T(2) - 6\,T(1))$

$= 5 \times 4 - 6 \times 2$

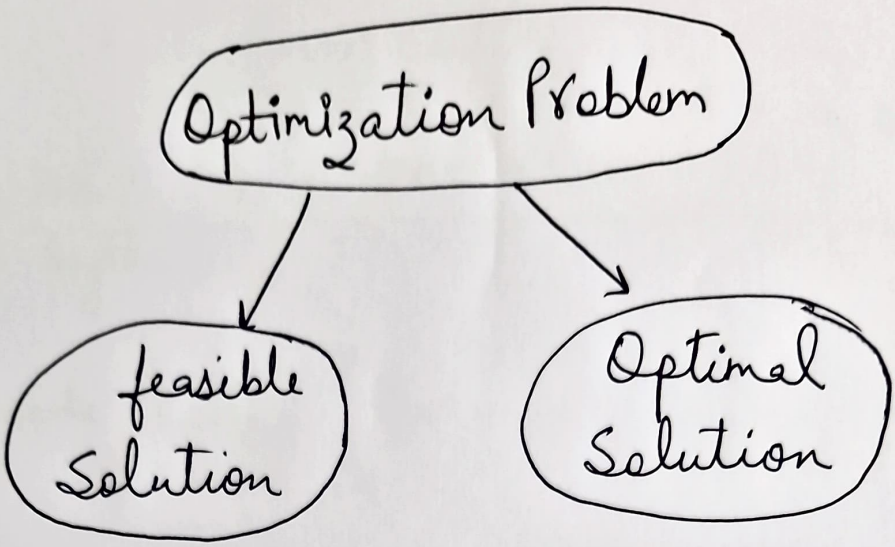
$= 8$

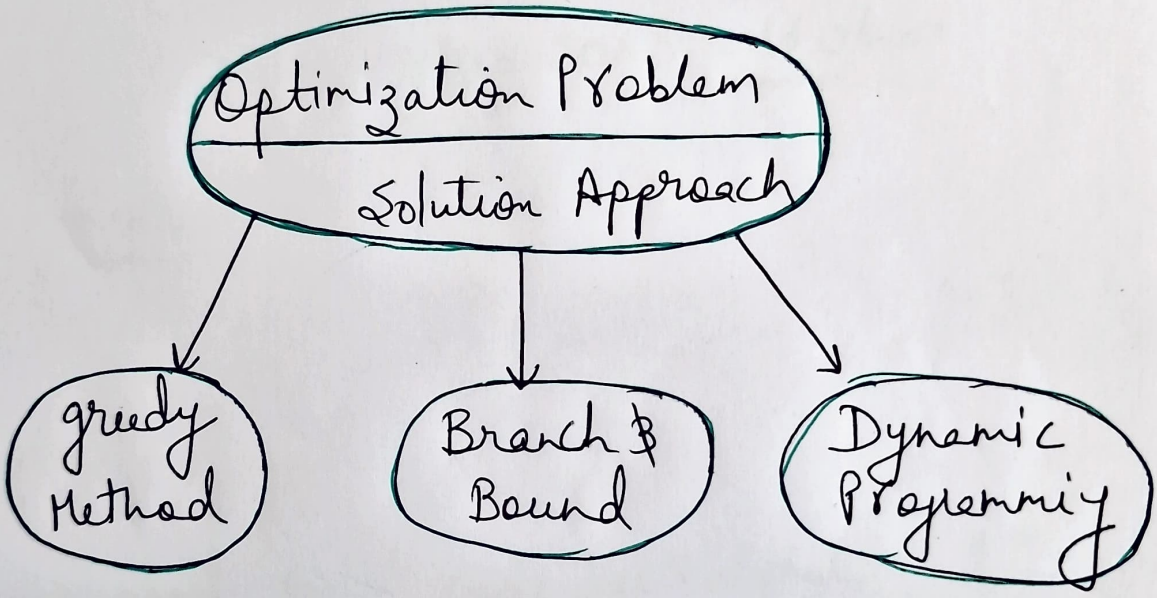
$- 2^0$

$- 2^1$

$- 2^2$

$\rightarrow 2^3$


Optimization Problem
feasible Solution
Optimal Solution


Optimization Problem
Solution Approach
greedy Method
Branch & Bound
Dynamic Programming

Huffman Coding

Message: BCC ABB DD A ECC BBAE DDCC } length = 20

Message length:

ASCII code (8 bits)

A ⇒ 65

B ⇒ 66

——— C, D, E ...

Message size = 20 × 8 = 160 bits

A → 01000001

B

C

D

E

Q Do we need ASCII code for 5 alphabets?

character	code (3 bits)
A	000
B	001
C	010
D	011
E	100
.	.

(Message size) = 20 × 3 = 60 bits

(Table size) = 5 × 8 + 5 × 3
= 55 bits

(Compression Ratio) = 160 / 115

→ Total = 60 + 55
= ~~70 bits~~ 115 bits

Character	frequency	
A	3	011
B	5	10
C	6	11
D	4	00
E	2	010
	20	Messy size $= \Sigma f_i d_i$

Messy size

$= 3\times3 + 5\times2 + 6\times2 + 4\times2 + 2\times3$

$= 9 + 10 + 12 + 8 + 6$

$= 45$ bits

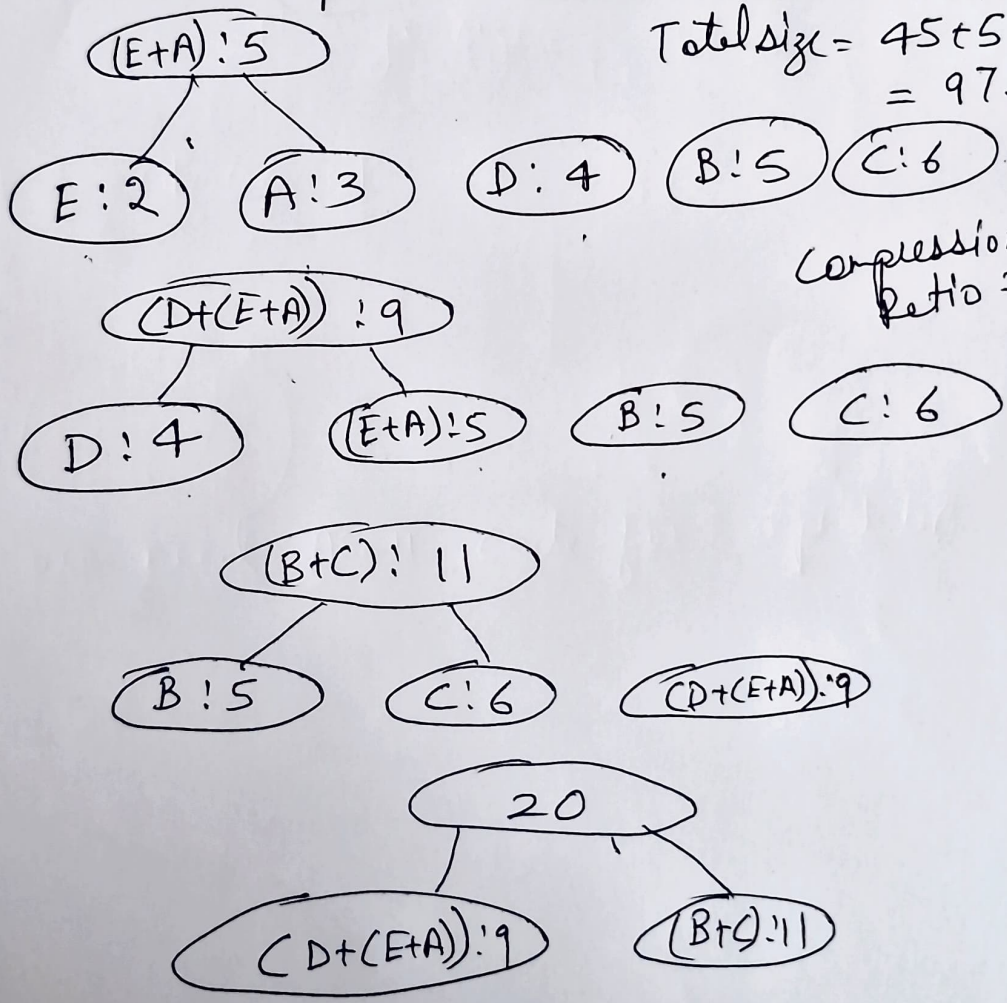
Table size $= 5\times8 + 3\times2 + 2\times3$

$= 52$ bits

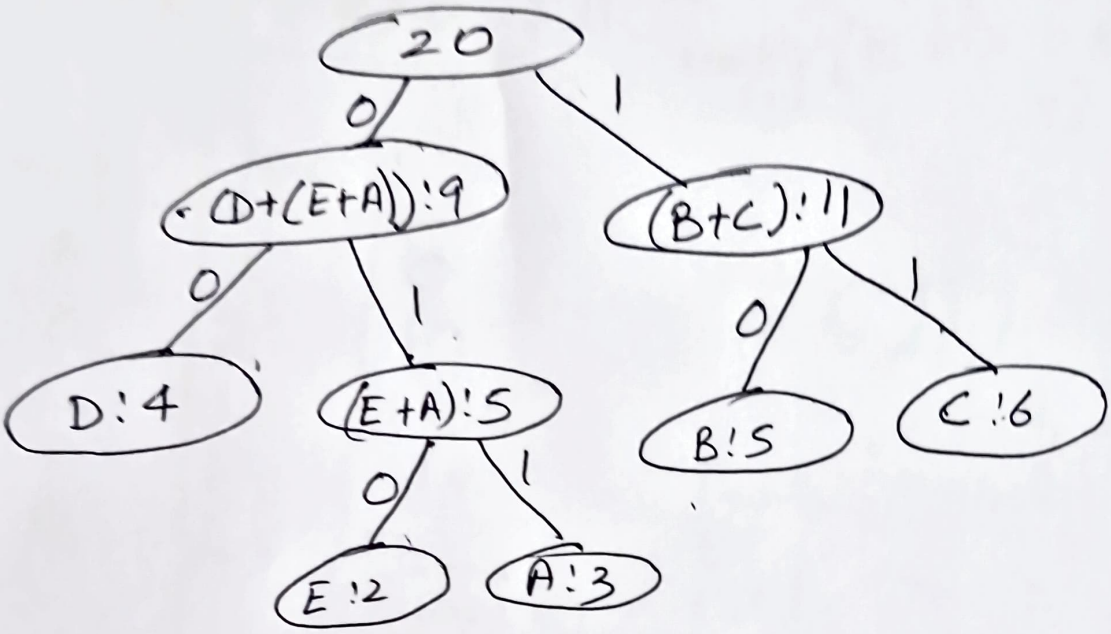
Total size $= 45 + 52$

$= 97$ bits

Compression ratio $= \frac{160}{97}$



Time Complexity $= O(n \log n)$.


20
0
1
(D+(E+A)):9
(B+C):11
0
1
0
1
D:4
(E+A):5
B:5
C:6
0
1
E:2
A:3

Q ABRACA DA BRA] 11

Message size (ASCII) = 11 × 8 = 88

Character	Freq.	
A	5	0
B	2	10
C	1	1100
D	1	1101
R	2	111

Message size = $\Sigma f_i d_i$

= 23 bits

Table = 54 bits

Size

Total = 77 bits

Compression Ration = 8/7

